

Phase 1 — Discovery & Requirements Lock

LineTwin · Accenture Innovation Challenge 2026 · Round 2 · Problem Track 4 "DigitalTwin.ai" Deliverable 2 of 3: Working Prototype

Purpose

Map every requirement in the Round 2 brief to a feature before any engine code exists, and fix the citation ledger so that no unverified number can enter the project later. Phase 1 produces no runnable code by design — it produces the constraints every later phase is checked against.

Starting position, stated honestly

The working directory contained reference material only: five journal PDFs, a patent, three internal notes, and the Round 1 problem statement. `git status` reported no repository. **No project code existed at the start of this phase.**

Toolchain verified before planning: `uv` 0.11.24, Node 24.16, git 2.39.2, 19 GB free disk. The system Python is 3.13, so `uv` will provision the pinned 3.12 interpreter itself — the pin is binding because `xgboost 3.4.1` requires ≥ 3.12 .

What Round 2 asks for

#	Deliverable	Owner
1	Detailed Business Proposal — problem framing, solution design, target users, business case and impact, phased roadmap, key risks with mitigations	Teammates
2	Working Prototype — a functional demonstration of the core mechanism, on illustrative or sample data	This repository
3	Pitch Presentation — presenting both proposal and prototype	Teammates

The brief states the prototype "does not need to be production-grade or use real enterprise data; a working proof-of-concept on illustrative or sample data is expected and encouraged."

Confirmed by reading both the Round 1 problem-statement PDF and the Round 2 brief in full: there is **no evaluation rubric, no scoring weights, no official metric, no leaderboard, and no technology mandate or ban**. Recorded in `REQUIREMENTS.md` §E so no self-imposed constraint is later mistaken for an external one.

Deliverables produced

Artefact	What it fixes
docs/REQUIREMENTS.md	All 7 real-world complexities and 6 solutioning areas traced to a feature, a module, and the evidence that will prove it
docs/CITATIONS.md	Three-part ledger: verified sources · forbidden numbers · our own numbers with mandatory qualifications
docs/DECISIONS.md	Identity, architecture, absorbed changes, scope cuts, contingency ladder, open items
docs/PRIOR_ART.md	The Accenture patent: what it claims, what we adopt, how we differ, and mandatory citation discipline
docs/tracker.html	Live progress tracker, published

Findings that changed the build

Twelve sources were read in full during this phase. Four produced changes that were not in the original build brief.

1 · A significance gate before naming any bottleneck. Kumbhar, Ng & Bandaru (2023) validate bottleneck claims with one-way ANOVA across stations followed by a Tukey–Kramer post-hoc test for unequal sample sizes. Adopted. Our detector now returns "*no significant bottleneck*" rather than always naming a station. This is a published, citable answer to the brief's warning that false alarms erode floor-level trust.

The same paper argues its Busy Ratio detector is preferable to the Active Period Method because active-period identification fluctuates week to week. That comparison is single-site with no ground truth and never establishes which detector was correct, so it does not overturn our method choice — but the volatility observation is real, and the significance test is the right response to it.

2 · Three-state value tagging. Detzner & Eigner (2018) require that zero, missing, and not-applicable remain three distinct states, never collapsed. Combined with the OBSERVED / INFERRED / SIMULATED tagging from the internal architecture note, every value in the twin now carries its provenance, a confidence, and a freshness. This directly prevents the failure mode of treating an estimate as a measurement — which matters more here than usual, because a third of our stations have no instrumentation at all.

The same paper supplies something more valuable: a published warning that **spurious correlations are amplified by a low failure base rate**, and that raising the significance threshold to compensate discards real signal. That is the citable justification for our calibration-and-baseline discipline, and it comes from the literature rather than from us asserting it.

3 • Transfer-delay timestamp realignment. US 12,353,197 B2 discloses computing contributing-factor timestamps as (*alert time – cumulative lag*) so upstream evidence is compared on the correct time frame. Adopted directly, with attribution. See below.

4 • Per-station rather than global thresholds. Fani et al. (2026) find per-workstation thresholds outperform a single global threshold. Applied wherever thresholds are used.

Prior art: a patent held by the judging organisation

LineTwin's defect-genealogy feature falls within the field of **US 12,353,197 B2**, *System contextualization of manufacturing plants*, assignee **Accenture Global Solutions Limited**, granted 8 July 2025.

This is declared rather than discovered. The patent claims graph-traversal root-cause attribution with transfer-delay realignment. We adopt the realignment and cite it. We differ in attribution: the patent chains fixed thresholds and produces a binary result with no disclosed false-positive rate; we carry a calibrated probability, publish the calibration curve, and gate bottleneck claims on a significance test.

Its contribution is *contextualisation* — getting messy plant data into a queryable graph. Ours is *inference and calibration* — what to do when that graph has holes. Complementary, not competing.

Citation discipline is mandatory and recorded in `PRIOR_ART.md` : a granted patent means a claim set was allowed over prior art, **not** that the method works. Always *discloses* or *claims*; never *demonstrates* or *validates*. The document contains no dataset, no evaluation, and no performance figure of any kind, so no number from it may be cited.

Data: simulated is not fabricated

Fabricating means inventing numbers and presenting them as measured. *Simulating* means a discrete-event queueing model computing telemetry from stated first principles — seed-reproducible, derived, and labelled as model output. The brief explicitly encourages the second.

A genuine search was run for real automotive assembly-line data rather than assuming none existed. The finding: **per-station working/starved/blocked state across a connected line topology is essentially not public**, because it is proprietary MES and PLC data. What exists nearby:

Dataset	Verdict
Bosch Production Line Performance	Real automotive, 51 stations — but features are anonymised with no released semantics, so no mapping to our features exists even in principle. Used for one published statistic only (~0.58 % defect prevalence, as our calibration target). Never downloaded, never trained on
Future Factories V2 (Univ. of South Carolina)	Real industrial assembly pipeline, 292 cycles, 21 cycle states. Closest real analogue — grounds cycle-time and state-duration parameters

Dataset	Verdict
PyScrew	Real industrial screw-driving series — grounds fastening-station variability, relevant to automotive torque stations
AI4I 2020 (UCI, CC BY 4.0)	Real, public, 10k rows — the offline benchmark model, proving the modelling capability on real data with real metrics
CarDA	Real car-door assembly, but RGB-D video and motion capture — wrong modality for a discrete-event twin

Consequence for the build: simulation parameters are calibrated against cited sources rather than chosen from thin air. Every value in `scenarios/line6.yaml` carries a `source:` field, or is stamped `synthetic – uncalibrated` with a note naming what would calibrate it.

Sources rejected

deep-research-report (3).md — a generic data-science-competition playbook assuming a leaderboard, an official numeric metric, GPU-scale deep learning, a 3–6 person team and a 12–24 week schedule. None of those exist in this challenge. Following it would have optimised for the wrong objective entirely.

Uncited performance thresholds in an internal predictive-techniques note (precision $\geq 98\%$, false alarm rate $\leq 0.5\%$, latency < 50 ms, 500-cycle trial). These have no provenance or benchmark in the source. Adopting them as achieved *or even as targets* would fabricate a standard. Excluded entirely; the staged-validation-gate narrative around them survives as a single roadmap sentence.

Thirteen individual figures from otherwise-good papers are named and banned in `CITATIONS.md` §2 — each real, each misleading in our context. The most dangerous are Mendi's 87.56 % (a *downtime* reduction, trivially misread as accuracy) and Palotai's ~\$1 M/year (self-caveated by its authors as an unverified extrapolation).

Exit criteria

Criterion	Status
Every brief complexity and solutioning area traced to a feature and its evidence	Met — <code>REQUIREMENTS.md</code> §A, §B
Citation ledger locked, forbidden figures named individually	Met — <code>CITATIONS.md</code> §2, 13 figures + 5 claim types
Identity, licence, station-count framing and positioning decided	Met — <code>DECISIONS.md</code> §1
Scope cuts recorded so later phases cannot quietly re-add them	Met — <code>DECISIONS.md</code> §4

Criterion	Status
Prior-art position documented with mandatory citation discipline	Met — <code>PRIOR_ART.md</code>
Tracker published	Met

Open items carried forward

None block Phase 2. Defaults are in force until overridden.

1. **Repo visibility** — local only; no remote push without explicit sign-off.
2. **Hosted deployment** — needs an account; deferred to Phase 10, falls back to a documented limitations entry.
3. `CITATION.cff` **authorship** — placeholder until teammate names are supplied.
4. **Fresh-machine clean-clone test** — unassigned. Should be run by someone other than the author, ideally on a machine without Python installed; any friction they hit is friction a judge hits.

Next

Phase 2 — Scaffold. Repository, pinned dependencies with ML and dev groups isolated from the server import path, CI on ubuntu and windows, and the frozen data contract that every later phase codes against. Exit gate: 60 valid fixture frames conforming to the frozen contract, CI green on both operating systems.